

74. Global Considerations in Kidney Disease_ Latin America · assets 1-9 / 10

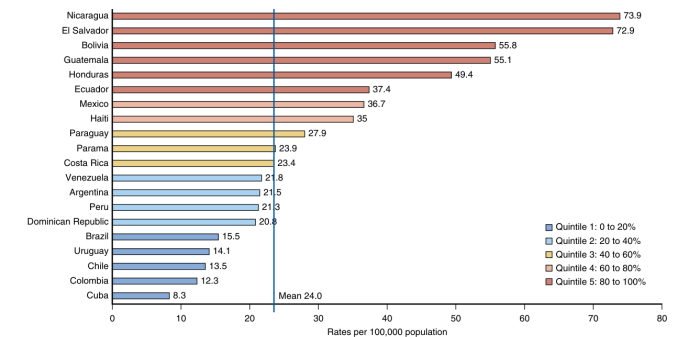


Fig. 74.1 Age-standardized death rates due to CKD in Latin America and the Latin Caribbean, per 100,000 population in 2019. (Reproduced from: Pan American Health Organization. The burden of kidney diseases in the Region of the Americas, 2000-2019. Washington, D.C.: PAHO; 2021. Available from: <https://www.paho.org/en/enlace/burden-kidney-diseases>; and accessed on 11th May 2025.)

Fig_74_1

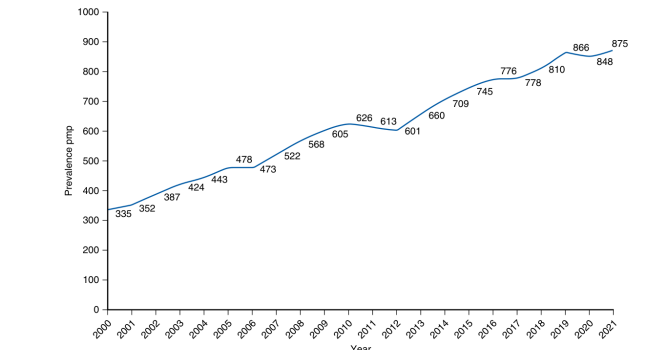


Fig. 74.4 Prevalence of treated end-stage kidney disease and kidney transplantation rates in Latin America and the Latin Caribbean, 2000-2021.

Fig_74_4

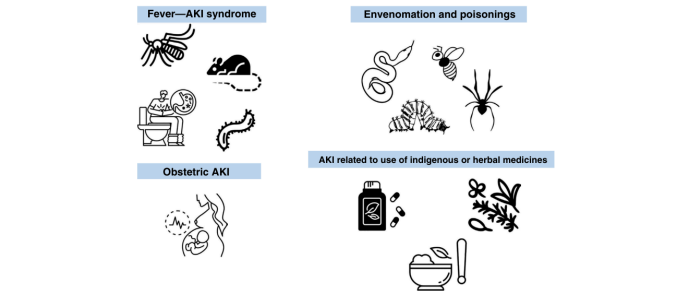


Fig. 74.7 Community-acquired acute kidney injury (AKI) in the tropics can be part of a constellation of manifestations, presenting as specific patterns of AKI syndromes, which help the clinician in deciding a diagnostic approach.

Fig_74_7

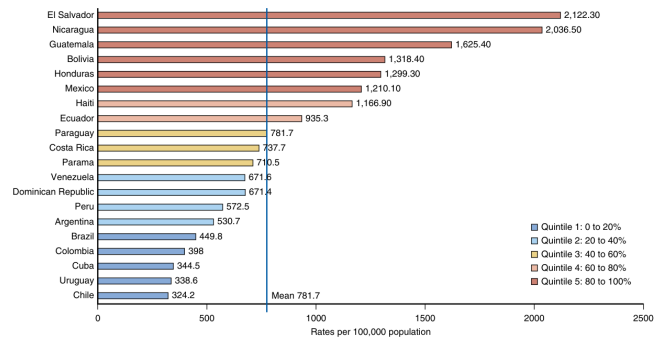


Fig. 74.2 Age-standardized disability-adjusted life years (DALYs) due to chronic kidney disease in Latin America and the Latin Caribbean, per 100,000 population in 2019. (Reproduced from: Pan American Health Organization. The burden of kidney diseases in the Region of the Americas, 2000-2019. Washington, D.C.: PAHO; 2021. Available from: <https://www.paho.org/en/enlace/burden-kidney-diseases> and accessed on 11th May 2025.)

Fig_74_2

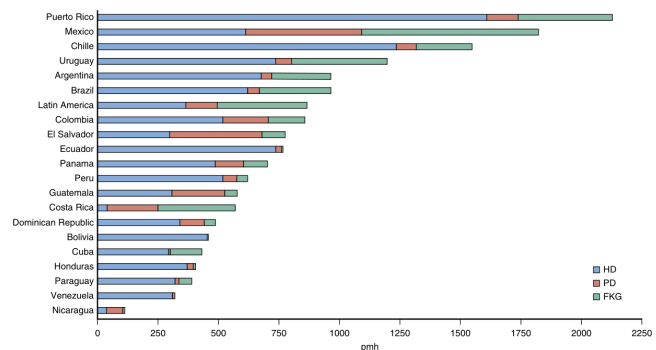


Fig. 74.5 Prevalence of treated end-stage kidney disease by treatment modality in Latin America and the Latin Caribbean, 2019. (From SLANH. Registro Latinoamericano de Diálisis y Trasplante. Reporte 2021. <https://slanh.net/registros/>; 2023. Accessed March 2023.)

Fig_74_5

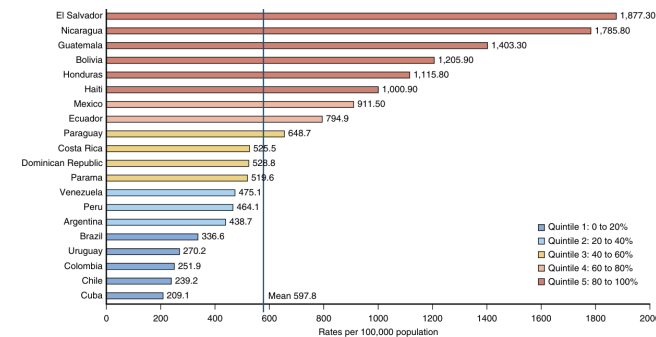


Fig. 74.3 Age-standardized years of life lost (YLLs) due to chronic kidney disease in Latin America and the Latin Caribbean, per 100,000 population in 2019. (Reproduced from: Pan American Health Organization. The burden of kidney diseases in the Region of the Americas, 2000-2019. Washington, D.C.: PAHO; 2021. Available from: <https://www.paho.org/en/enlace/burden-kidney-diseases> and accessed on 11th May 2025.)

Fig_74_3

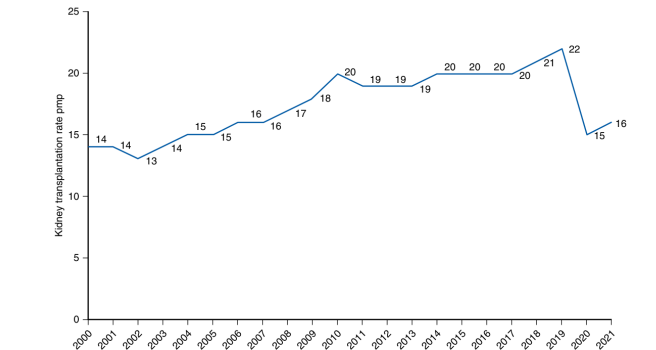


Fig. 74.6 Kidney transplantation rates in Latin America and the Latin Caribbean (2000-2021).

Fig_74_6

Table 74.2 Mortality, DALYs, and YLLs Attributable to CKD in Latin America and Latin Caribbean			
Country	Deaths Per 100,000 Population (95% UI)	DALYs Per 100,000 Population (95% UI)	YLL Per 100,000 Population (95% UI)
Central America, Mexico and the Latin Caribbean			
Costa Rica	23.4 (12.4-38.1)	737 (316-1159)	525 (267-899)
Cuba	8.3 (4.5-13.6)	344 (161-527)	209 (106-348)
Dominican Republic	20.8 (8.8-37.9)	671 (211-1130)	523 (238-979)
Guatemala	55.1 (29.9-80.6)	1626 (845-2605)	1400 (733-2361)
Haiti	35.0 (15.4-70.4)	1166 (390-1943)	1000 (425-2026)
Honduras	49.4 (24.6-88.3)	1299 (389-2210)	1115 (534-2029)
El Salvador	72.9 (37.4-124.3)	2122 (693-3951)	1877 (925-3288)
Mexico	36.7 (21.4-55.9)	1210 (656-1763)	911 (514-1415)
Nicaragua	73.9 (40.2-120.2)	2036 (836-3239)	1785 (939-2959)
Panama	23.9 (12.8-38.6)	710 (303-1117)	519 (267-881)
Andean Region			
Bolivia	12.3 (6.5-20.5)	398 (174-621)	251 (129-430)
Ecuador	37.4 (20.4-60.7)	935 (301-1479)	794 (417-1321)
Peru	21.3 (10.7-36.9)	572 (201-943)	464 (225-824)
Venezuela	21.8 (11.5-36.2)	671 (286-1056)	475 (241-811)
Southern Cone and Brazil			
Argentina	21.5 (12.3-33.6)	530 (256-805)	438 (241-700)
Brazil	15.5 (9.3-22.5)	449 (267-631)	336 (199-496)
Chile	13.5 (7.8-21.0)	324 (163-484)	239 (134-379)
Paraguay	27.9 (14.6-46.8)	781 (299-1263)	648 (327-1115)
Uruguay	14.1 (8.8-21.0)	338 (186-497)	270 (166-407)

CKD, Chronic kidney disease; DALY, disability-adjusted life years; UI, uncertainty interval; YLL, years of life lost. Reproduced from: Pan American Health Organization. The burden of kidney diseases in the Region of the Americas, 2000-2019. Washington, D.C.: PAHO; 2021. Available from: <https://www.paho.org/en/enlace/burden-kidney-diseases> and accessed on 11th May 2025.

Table_74_2

Table 74.3 Incidence and Prevalence of Treated End-Stage Kidney Disease by Treatment Modality in Latin America and Latin Caribbean, 2019									
Country	Population	Incidence Rate (pmp)	HD	PD	Total DALYs	% on PD	FKG	Total Prevalence (pmp)	Kidney Transplant Rate (pmp)
Central America, Mexico, and Latin Caribbean									
Costa Rica	5,123,105	38	40	209	249	83.9	318	567	15
Cuba	11,300,698	158	293	6	299	2.0	131	430	15
Dominican Republic	10,959,664	221	340	98	438	22.3	47	485	5
Guatemala	17,362,718	140	304	221	525	42.0	51	575	8
Haiti	11,306,801	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Honduras	10,121,763	96	370	22	392	5.6	13	405	0
El Salvador	6,262,731	677	297	380	677	56.1	89	776	6
Mexico	125,898,302	530	611	483	1094	44.1	729	1823	62
Nicaragua	6,755,895	31	35	65	100	65.0	11	111	2
Panama	4,294,396	181	488	113	601	18.8	100	701	8
Puerto Rico	3,193,694	419	1607	130	1737	7.4	382	2129	18
Andean Region									
Bolivia	11,936,160	114	452	2	454	0.44	3	457	2
Colombia	50,930,662	103	516	185	701	26.3	157	858	19
Ecuador	17,588,595	8	735	21	756	2.7	12	768	13
Peru	30,364,796	62	915	57	972	9.9	40	818	3
Venezuela	28,490,453	96	310	10	320	3.1	0	320	1
Southern Cone and Brazil									
Argentina	45,036,032	163	674	46	720	6.3	243	963	35
Brazil	213,196,304	219	616	47	665	7.0	299	963	30
Chile	19,320,315	204	1236	81	1317	6.1	233	1550	22
Paraguay	6,618,695	36	317	16	353	4.8	54	387	4
Uruguay	3,429,086	185	734	62	796	7.7	398	1194	42

FKG, Functioning kidney graft; HD, hemodialysis. Modified from: Luviano R, Corroto L, Gonzalez-Badst M, et al. The Latin American Dialysis and Renal Transplantation Registry: report 2019. *On Kidney J.* 2022;15(3):425-431.

Table_74_3

Table 74.4 Causes of Acute Kidney Injury in the Tropics	
Vector-borne infections	Malaria (Plasmodium falciparum, Plasmodium vivax, Plasmodium knowlesi, transmitted by Anopheles mosquitoes), Dengue fever (dengue virus, transmitted by Aedes triseriatus), South Asian Chikungunya (chikungunya virus, transmitted by Aedes albopictus), Rift Valley fever (RVF virus, transmitted by Aedes or Culex mosquitoes), Leishmaniasis (Leishmania spp.), Hepatitis, HIV (via oral exposure), Hemorrhagic fever with renal syndrome (Puumala and Hantaan viruses), Zoonoses
Diarrheal diseases	Viral diarrhea (rotavirus, Norwalk agent), Enterocolitis, or enteroinvasive Escherichia coli, Bacillary dysentery (Shigella)
Other infections	Cholera, Malaria (Plasmodium falciparum), Typhoid (Salmonella typhi), Giardiasis (Giardia lamblia), Lepidopteran (Lepidoptera) pneumoniae, Human immunodeficiency virus
Plant and fungal toxins	Herbicides, Insecticides, Dried leaves, Moringa, Sesame, Sorghum, and other edible-rich plants, Plant-derived toxins used as insecticides and for tea
Animal poisons	Snakebites, Wasps, bees, and bee stings, Spider bite, Jellyfish sting
Chemical nephrotoxins	Scorpion sting, Carbamates or herbicides, Organophosphates, Paracetamol/acetaminophen, Ethylene glycol, Chlorine, Sulfuric acid
Environmental factors	Heat stroke, Dehydration, Heat-related illnesses
Obstetric complications	Pre-eclampsia, Eclampsia, and postpartum hemorrhage

Table_74_4